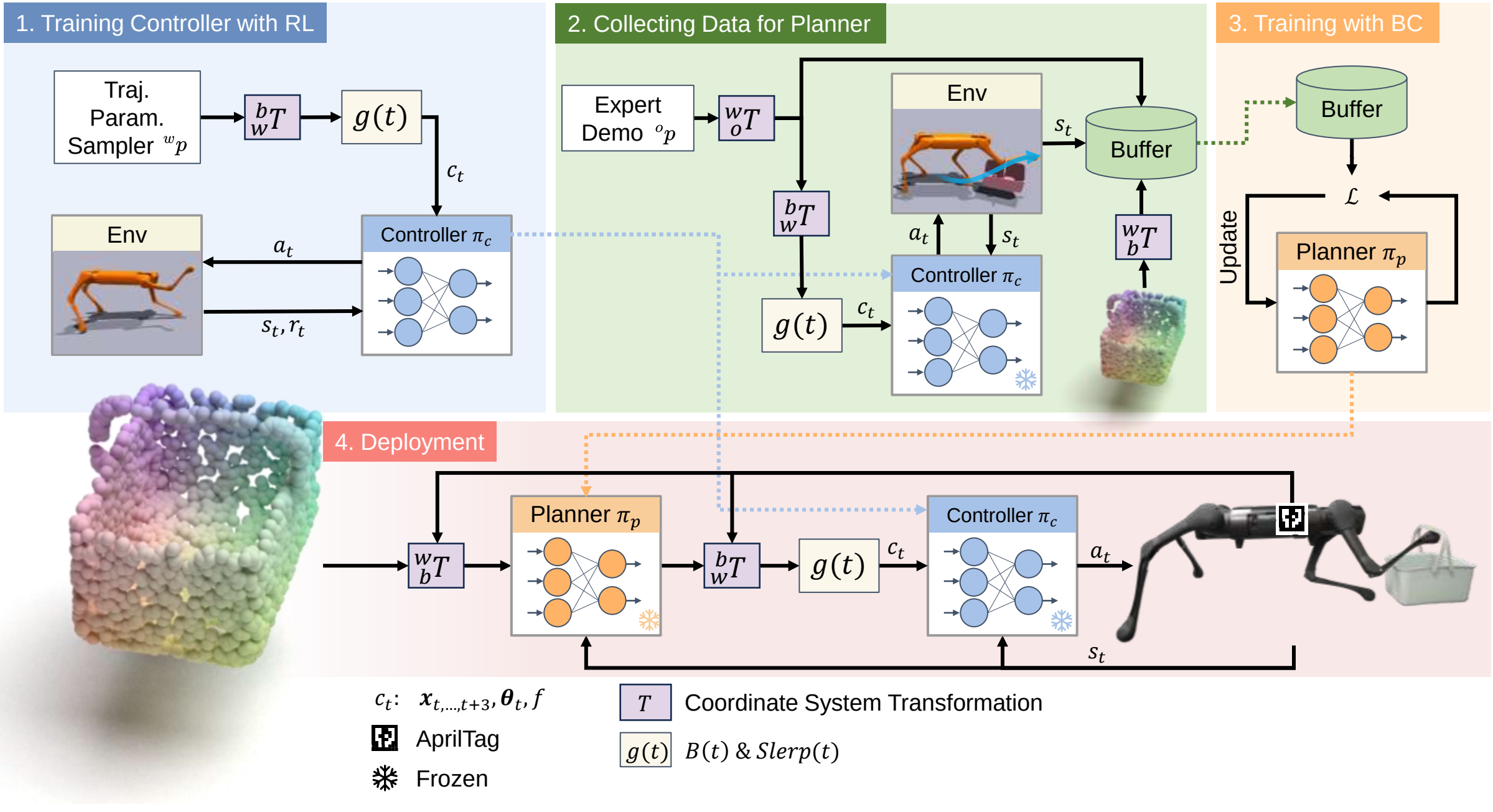
1. Training Controller with RL
Traj. Param. Sampler ${}^{w}p$
${}^{b}_{w}T$
$g(t)$
$c_t$
Env
Controller $\pi_c$
$a_t$
$s_t, r_t$
2. Collecting Data for Planner
Expert Demo ${}^{o}p$
${}^{w}_{o}T$
Env
$s_t$
Buffer
${}^{b}_{w}T$
$a_t$
$s_t$
${}^{w}_{b}T$
$g(t)$
$c_t$
Controller $\pi_c$
3. Training with BC
Buffer
$\mathcal{L}$
Update
Planner $\pi_p$
4. Deployment
Planner $\pi_p$
${}^{w}_{b}T$
${}^{b}_{w}T$
$g(t)$
$c_t$
Controller $\pi_c$
$a_t$
$s_t$
$c_t$: $\boldsymbol{x}_{t,\ldots,t+3}, \boldsymbol{\theta}_t, f$
AprilTag
Frozen
$T$ Coordinate System Transformation
$g(t)$ $B(t)$ & $Slerp(t)$